

# Manikantan (Mani) Srinivasan

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[LinkedIn](#)

[Portfolio](#)

[GitHub](#)

## EDUCATION

**Northeastern University**, Boston, Massachusetts

Sep 2023 – Aug 2025

**Khoury College of Computer Sciences**

(expected)

Master of Science in Artificial Intelligence- GPA: 3.5

*Coursework: NLP, AI in Human Computer Interaction, LLM, Machine Learning, Data Science, Unsupervised ML*

*Teaching Assistant for CS6220 Data Mining – teaching graduate-level data mining techniques.*

**UPES**, Dehradun, Uttarakhand (India)

Aug 2019 – May 2023

Bachelor of Technology in Computer Science (specialization in AI&ML) – GPA: 8.3

*Coursework: Machine Learning, Algorithms, Deep Learning, Database Management, Software Engineering*

## SKILLS

**Languages:**

Python, C++, Java, C, R, HTML, CSS, Bash, MATLAB, Dart, Swift

**Software Development Tools**

Git, GitLab, Jira, Docker, Kubernetes, Xcode, Tableau, Snowflake

**Cloud and Databases**

GCP, AWS EC2, SageMaker, S3, ArangoDB, MySQL, MongoDB, Neo4j, Redis, Flutter,

**Libraries and Frameworks:**

PyTorch, HuggingFace, TensorFlow, Django, LangChain, Pandas, Sci-kit Learn, Matplotlib

**Domain Skills**

Statistical Modelling, Deep Learning, LLM Fine Tuning, RAG, AI Agents, Data Analytics

## PROFESSIONAL EXPERIENCE

[SignoFi](#), Raleigh, USA

Sep 2024 – Jan 2025

**AI Research Engineer Intern**

- Pioneered an Agentic AI application leveraging Llama 70B and LangChain to retrieve balance sheet data from NoSQL databases, enabling interactive queries and computational analysis for over 50,000 public companies.
- Leveraged RAG and Finetuning with adaptive chunking strategies to ground Llama 70B and 90B, boosting the Faithfulness score to 0.84 (84%) and enhancing contextual precision, recall, and relevancy (92%) in factual question answering.
- Architected a custom workflow on Google Cloud Platform using virtual machines, Vertex AI, GCP Buckets, and multi-core GPUs to optimize operations at a portfolio management firm, achieving up to 20% in cost savings.

[KEK](#), Tsukuba, Ibaraki (Japan)

Oct 2022 – May 2023

**Researcher – Distributed Computing (International Research Lab)**

- Collaborated on the Prestigious [BELLE-2](#) particle acceleration experiment as a student researcher. The collaboration involved 5+ world famous laboratories like Brookhaven, CERN, KEK and 127+ institutions like Carnegie Mellon.
- Engineered innovative software tools that empowered physicists to execute experiments on over 10 PB of data across global data centers.
- Streamlined 50+ issues on GitLab and JIRA by overhauling complex codebases and partnering with engineers worldwide.

[Jio- Reliance Industries](#), Mumbai, Maharashtra (India)

Jun 2022 – Aug 2022

**Software Engineer Intern – (Center of Excellence- AI & ML)**

- Engineered the BrainOS platform integrating 7+ verticals for Reliance Industries, utilizing knowledge graphs and predictive ML across products like JioTV, Hydrocarbons, and JioMart, achieving 86% accuracy and 0.79 precision.
- Developed robust knowledge graphs by preprocessing raw data and implemented 100+ CRUD API functions to efficiently query extensive NoSQL databases, utilizing debugging tools to analyze and refine 1000+ functions.
- Facilitated the development of a beta version of user-user collaborative filtering for product recommendations using Neo4j and custom graph functions as part of a cross-functional effort, helping drive a 10% revenue boost for Jio Platforms.

## PROJECTS

[BlindSight](#): AI-Powered Voice-Driven File Navigation for the Visually Impaired

Aug 2024 – Dec 2024

- Developed an **AI-powered voice navigation system**, enabling visually impaired users to perform file management and document editing tasks with 95% speech recognition accuracy using Open-AI's C++ implementation of **Whisper**.
- Incorporated **LLaMA-70B** with tailored in-context learning (ICL) to deliver task-specific capabilities, such as document summarization and advanced text editing, achieving 85% accuracy in real-world testing scenarios.
- Designed a multi-threaded architecture that reduced task execution latency by 30%.

[ABB](#): A RL powered Game with sentiment analysis-based character movement

Sept 2023 – Dec 2023

- Crafted a 10x10 grid-based game employing value iteration and sentiment analysis to steer a character via creative text input toward two goal states while optimizing points, featuring a custom reward function.

## ACHIEVEMENTS

- Selected (from a pool of 500 students) for the prestigious BELLE-2 Particle Accelerator experiment, a Nobel Prize-winning endeavor. - <https://www.belle2.org>